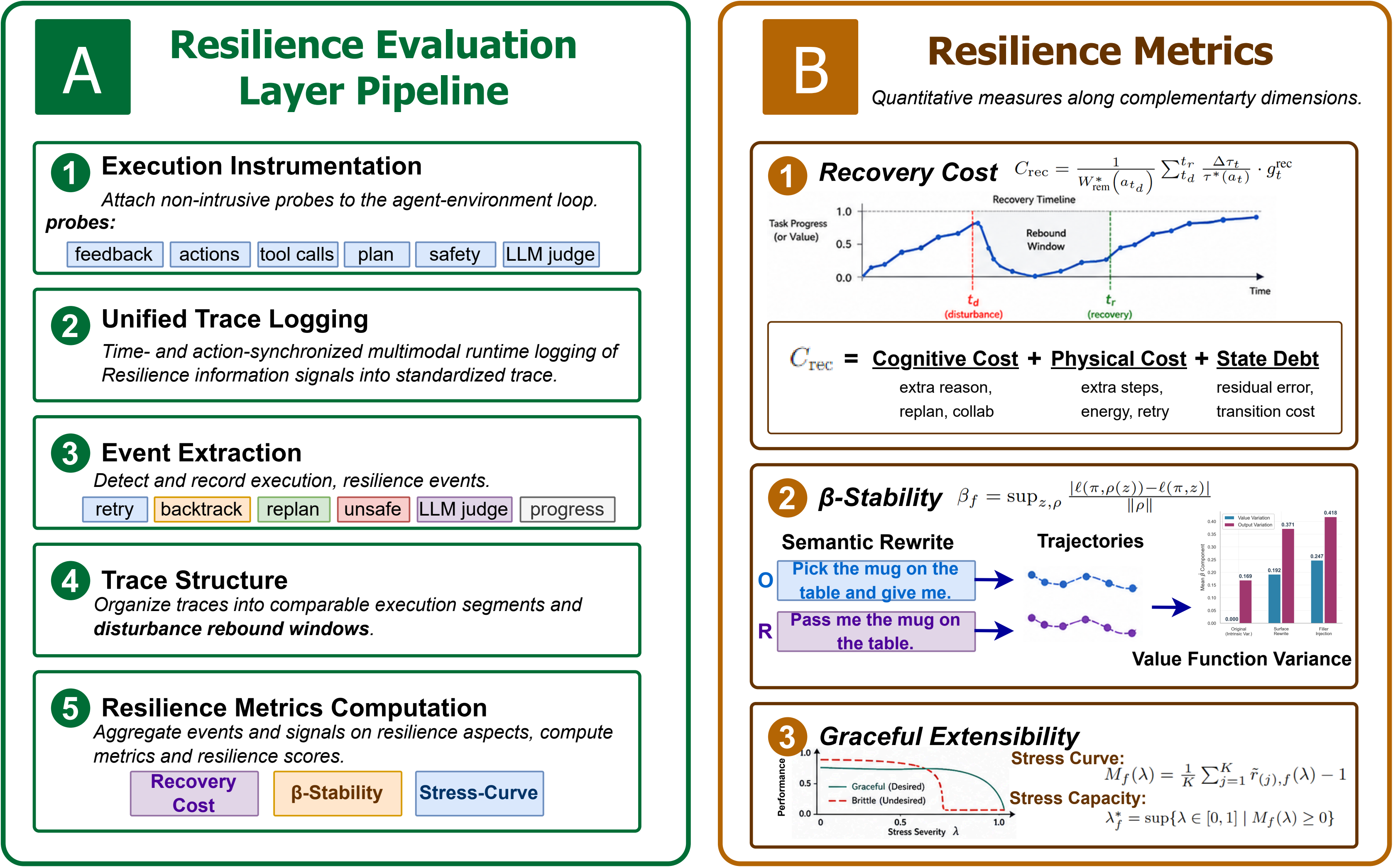

A
Resilience Evaluation Layer Pipeline
1 Execution Instrumentation
Attach non-intrusive probes to the agent-environment loop.
probes:
feedback
actions
tool calls
plan
safety
LLM judge
2 Unified Trace Logging
Time- and action-synchronized multimodal runtime logging of Resilience information signals into standardized trace.
3 Event Extraction
Detect and record execution, resilience events.
retry
backtrack
replan
unsafe
LLM judge
progress
4 Trace Structure
Organize traces into comparable execution segments and disturbance rebound windows.
5 Resilience Metrics Computation
Aggregate events and signals on resilience aspects, compute metrics and resilience scores.
Recovery Cost
β-Stability
Stress-Curve
B
Resilience Metrics
Quantitative measures along complementarty dimensions.
1 Recovery Cost $C_{\mathrm{rec}} = \frac{1}{W^*_{\mathrm{rem}}(a_{t_d})} \sum_{t_d}^{t_r} \frac{\Delta\tau_t}{\tau^*(a_t)} \cdot g_t^{\mathrm{rec}}$
Recovery Timeline
Task Progress (or Value)
1.0
0.5
0.0
Rebound Window
Time
t_d (disturbance)
t_r (recovery)
C_{rec} = Cognitive Cost + Physical Cost + State Debt
extra reason, replan, collab
extra steps, energy, retry
residual error, transition cost
2 β-Stability $\beta_f = \sup_{z,\rho} \frac{|\ell(\pi,\rho(z)) - \ell(\pi,z)|}{\|\rho\|}$
Semantic Rewrite
Trajectories
O
Pick the mug on the table and give me.
R
Pass me the mug on the table.
Value Variation
Output Variation
0.000
0.169
0.192
0.371
0.247
0.418
Value Function Variance
3 Graceful Extensibility
Performance
1.0
0.5
0.0
Graceful (Desired)
Brittle (Undesired)
0
0.5
1.0
Stress Severity λ
Stress Curve:
$M_f(\lambda) = \frac{1}{K} \sum_{j=1}^{K} \tilde{r}_{(j),f}(\lambda) - 1$
Stress Capacity:
$\lambda_f^* = \sup\{\lambda \in [0,1] \mid M_f(\lambda) \geq 0\}$